

SQL PROJECT LIFESTYLE

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PROJECT DESCRIPTION

This presentation is based on the analysis of three datasets:

- 1 - Global Country Information Dataset
- 2 - Global Data on Sustainable Energy
- 3 - SDG Index from 2000 to 2022

The project analyzes global performance in terms of sustainability, CO2 emissions, and socio-economic conditions. Through various SQL queries, it identifies countries with the highest and lowest scores in the SDG Index (Sustainable Development Goals Index), trends in CO2 emissions over time, and access to electricity. It also examines the correlation between sustainable development and environmental impact, with a particular focus on how these factors affect quality of life.

The project will be implemented using the PostgreSQL language and PgAdmin 4 software.



DATA CLEANING, DATABASE CREATION AND TABLE SETUP

country_2023

```
CREATE TABLE country_2023 (  
  country VARCHAR(255),  
  abbreviation VARCHAR(10),  
  birth_rate DECIMAL(10,2),  
  co2_emission DECIMAL(15,2),  
  gdp DECIMAL(20,2),  
  gross_primary_education_enrollment_percent DECIMAL(10,2),  
  gross_tertiary_education_enrollment_percent DECIMAL(10,2),  
  infant_mortality DECIMAL(10,2),  
  life_expectancy DECIMAL(10,2),  
  minimum_wage DECIMAL(15,2),  
  physicians_per_thousand DECIMAL(10,2),  
  population DECIMAL(20,2),  
  population_labour_force_percent DECIMAL(10,2),  
  unemployment_rate DECIMAL(10,2)  
);
```

DATA CLEANING, DATABASE CREATION AND TABLE SETUP

sustainable_energy

```
CREATE TABLE sustainable_energy (  
  entity VARCHAR(255),  
  year VARCHAR(10),  
  access_to_electricity_percent_of_population DECIMAL(10,2),  
  gdp_per_capita DECIMAL(20,2),  
  renewable_electricity_capacity_per_capita DECIMAL(15,2),  
  co2_emissions_kt_by_country DECIMAL(20,2),  
  renewable_energy_share_totalfinalenergy_percent DECIMAL(10,2),  
  electricity_from_renewables_twh DECIMAL(20,2),  
  low_carbon_electricity_percent DECIMAL(10,2),  
  renewables_percent_equivalent_primary_energy DECIMAL(10,2)  
);
```

sdg_index_2000_2022

```
CREATE TABLE sdg_index_2000_2022 (  
  country_code VARCHAR(10),  
  country VARCHAR(255),  
  year VARCHAR(10),  
  sdg_index_score DECIMAL(10,2),  
  goal_1_score DECIMAL(10,2),  
  goal_2_score DECIMAL(10,2),  
  goal_3_score DECIMAL(10,2),  
  goal_4_score DECIMAL(10,2),  
  goal_5_score DECIMAL(10,2),  
  goal_6_score DECIMAL(10,2),  
  goal_7_score DECIMAL(10,2),  
  goal_8_score DECIMAL(10,2),  
  goal_9_score DECIMAL(10,2),  
  goal_10_score DECIMAL(10,2),  
  goal_11_score DECIMAL(10,2),  
  goal_12_score DECIMAL(10,2),  
  goal_13_score DECIMAL(10,2),  
  goal_14_score DECIMAL(10,2),  
  goal_15_score DECIMAL(10,2),  
  goal_16_score DECIMAL(10,2),  
  goal_17_score DECIMAL(10,2)  
);
```

DATA ANALYSIS

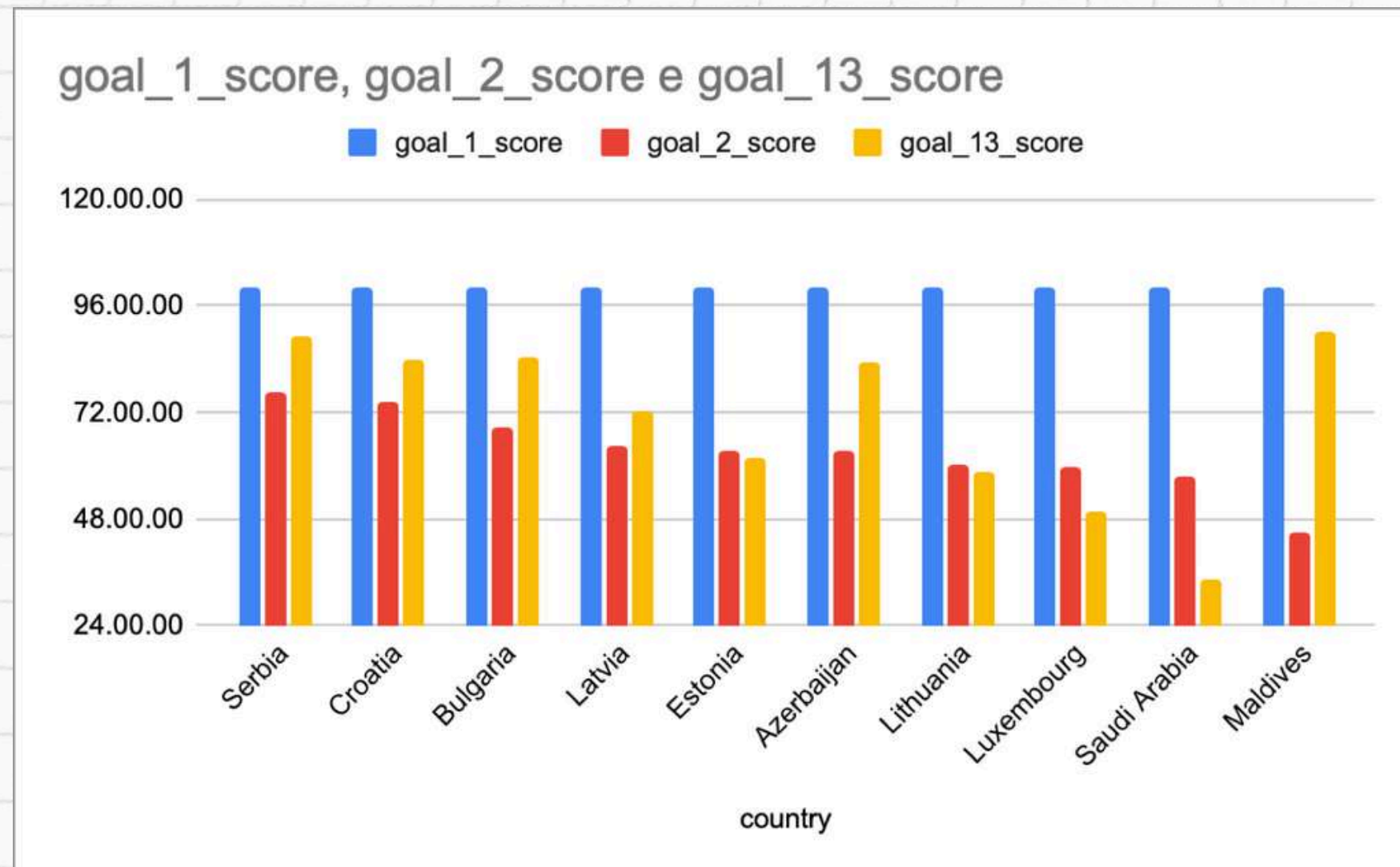
WHICH COUNTRIES HAVE THE HIGHEST SDG INDEX 2022 FOR GOALS 1, 2, AND 13?

```
SELECT country, goal_1_score, goal_2_score, goal_13_score
FROM sdg_index_2000_2022
WHERE year = '2022'
ORDER BY goal_1_score DESC, goal_2_score DESC, goal_13_score DESC
LIMIT 10;
```

The listed countries are Serbia, Croatia, Bulgaria, Latvia, Estonia, Azerbaijan, Lithuania, Luxembourg, Saudi Arabia, and the Maldives. All countries achieve a perfect score of 100 for **Goal 1 (poverty)**, but none reach 100 for **Goal 2 (hunger)**.

Scores for **Goal 2** vary significantly, with the highest achieved by Serbia (75.80) and the lowest by the Maldives (44.60).

Regarding **Goal 13 (climate action)**, the Maldives achieve the highest score (89.80), followed by Serbia (88.50).



WHICH COUNTRIES HAVE THE WORST SDG INDEX 2022 FOR GOALS 1, 2, AND 13?

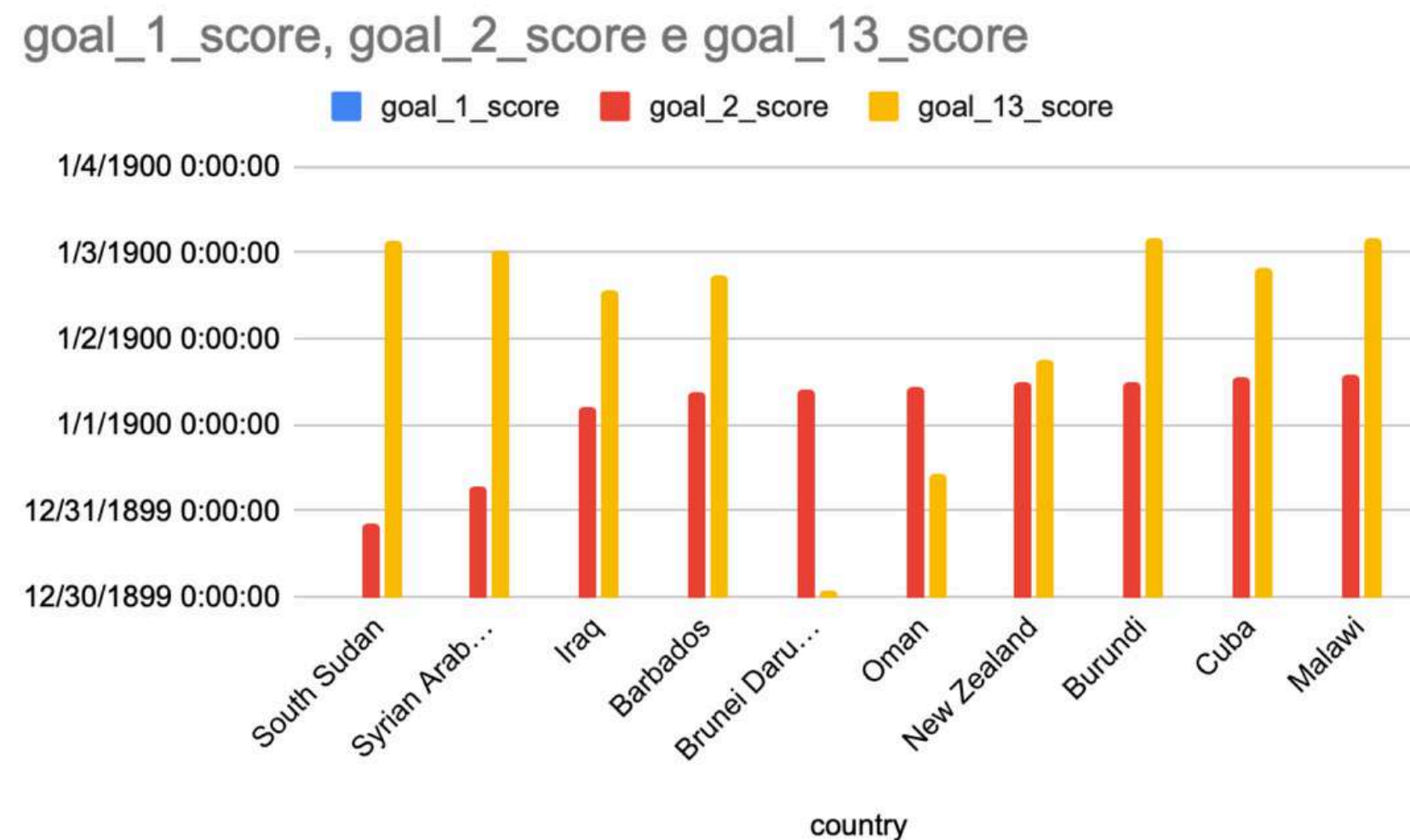
```
SELECT country, goal_1_score, goal_2_score, goal_13_score
FROM sdg_index_2000_2022
WHERE year = '2022'
ORDER BY goal_1_score ASC, goal_2_score ASC, goal_13_score ASC
LIMIT 10;
```

The countries listed are South Sudan, Syria, Iraq, Barbados, Brunei, Oman, New Zealand, Burundi, Cuba, and Malawi.

All these countries score 0 for **Goal 1 (poverty)**, indicating a severe crisis in this area.

Regarding **Goal 2 (hunger)**, scores range from the lowest (19.80) in South Sudan to the highest (61.70) in Malawi.

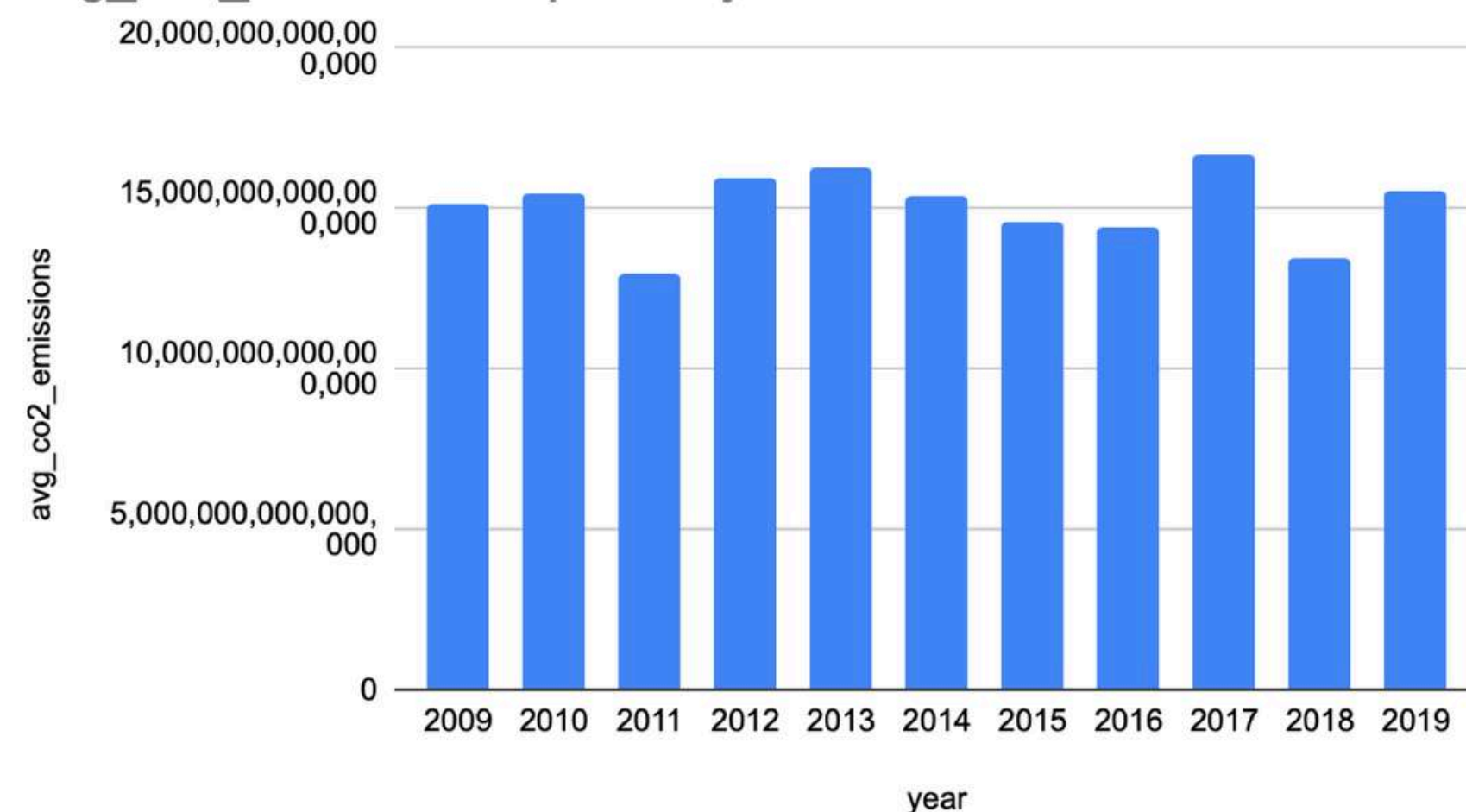
For **Goal 13 (climate action)**, scores vary drastically, with some countries like Brunei scoring only 1.30, while South Sudan records a surprisingly high score of 99.40.



GLOBAL CO2 EMISSIONS TREND OVER TIME (2009–2019)

```
SELECT year, AVG(co2_emissions_kt_by_country) AS avg_co2_emissions
FROM sustainable_energy
WHERE year BETWEEN '2009' AND '2019'
GROUP BY year
ORDER BY year;
```

avg_co2_emissions rispetto a year



The analysis of CO2 emissions from **2009** to **2019** shows a general upward trend, with significant fluctuations.

After a **peak in 2013**, emissions slightly **decreased until 2016**, then started rising again.

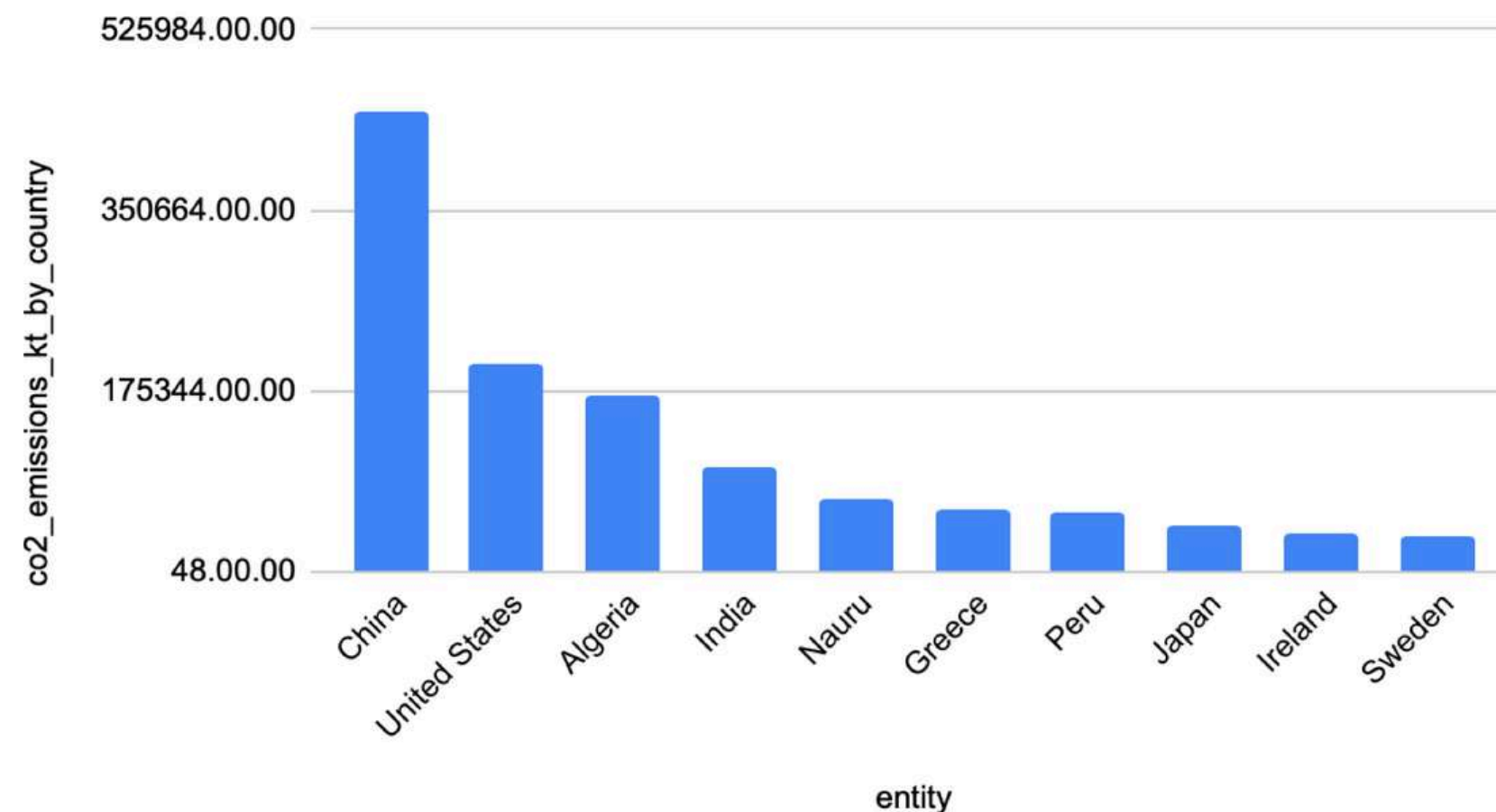
These variations are influenced by economic factors, energy policies, and natural events. A continuous global commitment is essential to reduce emissions and constantly monitor data to adapt environmental sustainability strategies.

WHICH COUNTRIES HAD THE HIGHEST CO2 EMISSIONS IN 2019?

```
SELECT entity, co2_emissions_kt_by_country
FROM sustainable_energy
WHERE year = '2019' AND co2_emissions_kt_by_country IS NOT NULL
ORDER BY co2_emissions_kt_by_country DESC
LIMIT 10;
```

- **China and the United States** are the largest CO2 emitters, contributing significantly to global emissions.
- **India and Russia** follow, with emissions exceeding one million kilotons.
- Other countries on the list, such as **Japan, Germany, and Iran**, also show high emission levels but are significantly lower than the top three.

co2_emissions_kt_by_country rispetto a entity



These data highlight the importance of targeted environmental policies in the countries with the highest emissions to reduce the global impact on climate change.

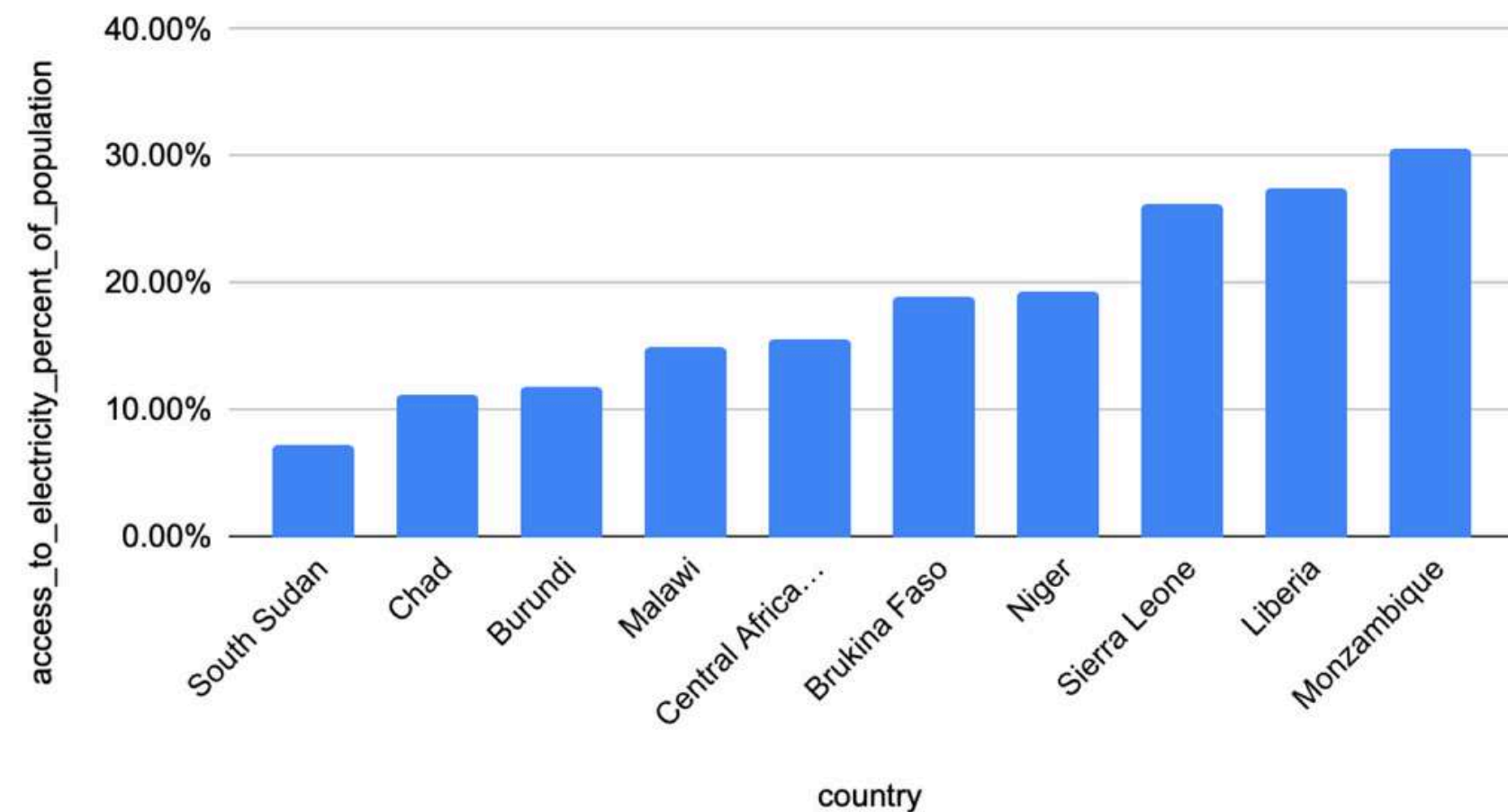
WHICH COUNTRIES HAD THE LOWEST ACCESS TO ELECTRICITY IN 2020?

```
SELECT entity AS country, access_to_electricity_percent_of_population
FROM sustainable_energy
WHERE year = '2020'
ORDER BY access_to_electricity_percent_of_population ASC
LIMIT 10;
```

This query lists the countries with the lowest percentage of access to electricity in 2020. The results show that **the poorest nations**, especially those in **Sub-Saharan Africa**, still face significant challenges in implementing electricity infrastructure for the entire population.

All listed countries have very low electricity access: **South Sudan, Chad, and Burundi** at around 10%, with **Mozambique** reaching a maximum of 30.60%.

access_to_electricity_percent_of_population rispetto a country



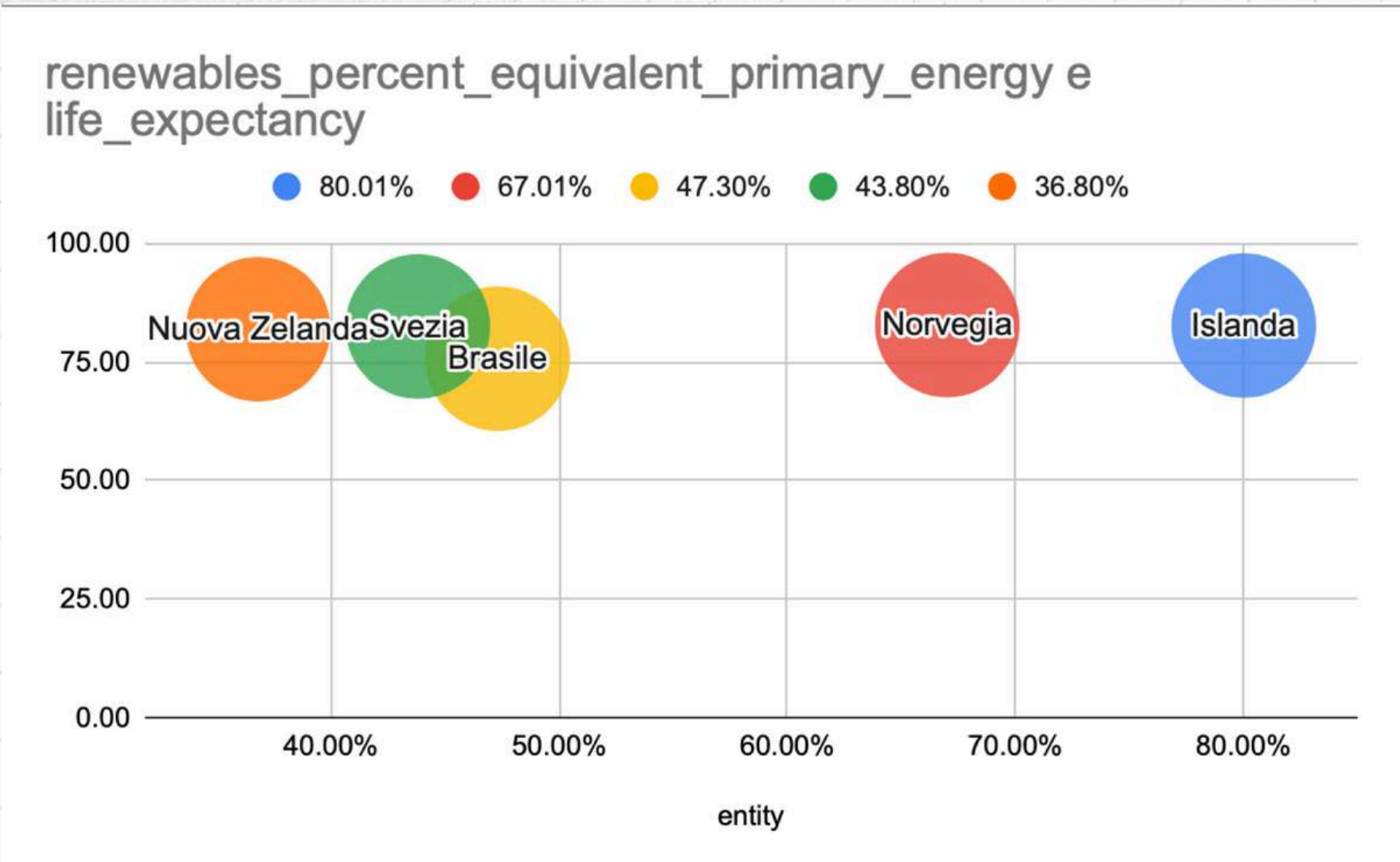
TOP 5 COUNTRIES BY CO2 EMISSIONS FROM 2016 TO 2019

```
WITH co2_emissions_ranking AS (  
  SELECT entity, year, co2_emissions_kt_by_country,  
         ROW_NUMBER() OVER (PARTITION BY year ORDER BY co2_emissions_kt_by_country DESC) AS rn  
  FROM sustainable_energy  
  WHERE year IN ('2019', '2018', '2017', '2016')  
         AND co2_emissions_kt_by_country IS NOT NULL  
)  
SELECT entity, year, co2_emissions_kt_by_country  
FROM co2_emissions_ranking  
WHERE rn <= 5  
ORDER BY year DESC, co2_emissions_kt_by_country DESC;
```

The results indicate that, between 2016 and 2019, the countries with the highest levels of CO2 emissions remained consistent. **China** ranks first, followed by the **United States, India, Japan**, and **Germany**. These five countries continue to be the largest global contributors to carbon dioxide emissions, reflecting their predominant position in terms of industrial development and energy consumption, despite international policies aimed at reducing environmental impact.

TOP 5 COUNTRIES WITH HIGH PERCENTAGES OF RENEWABLE ENERGY IN 2019 AND A LIFE EXPECTANCY OVER 75 YEARS

```
SELECT DISTINCT se.entity, se.year, se.renewables_percent_equivalent_primary_energy, c.life_expectancy
FROM sustainable_energy se
JOIN country_2023 c
  ON se.entity = c.country
WHERE se.year = '2019'
  AND se.renewables_percent_equivalent_primary_energy IS NOT NULL
  AND c.life_expectancy > 75
ORDER BY se.renewables_percent_equivalent_primary_energy DESC
LIMIT 5;
```



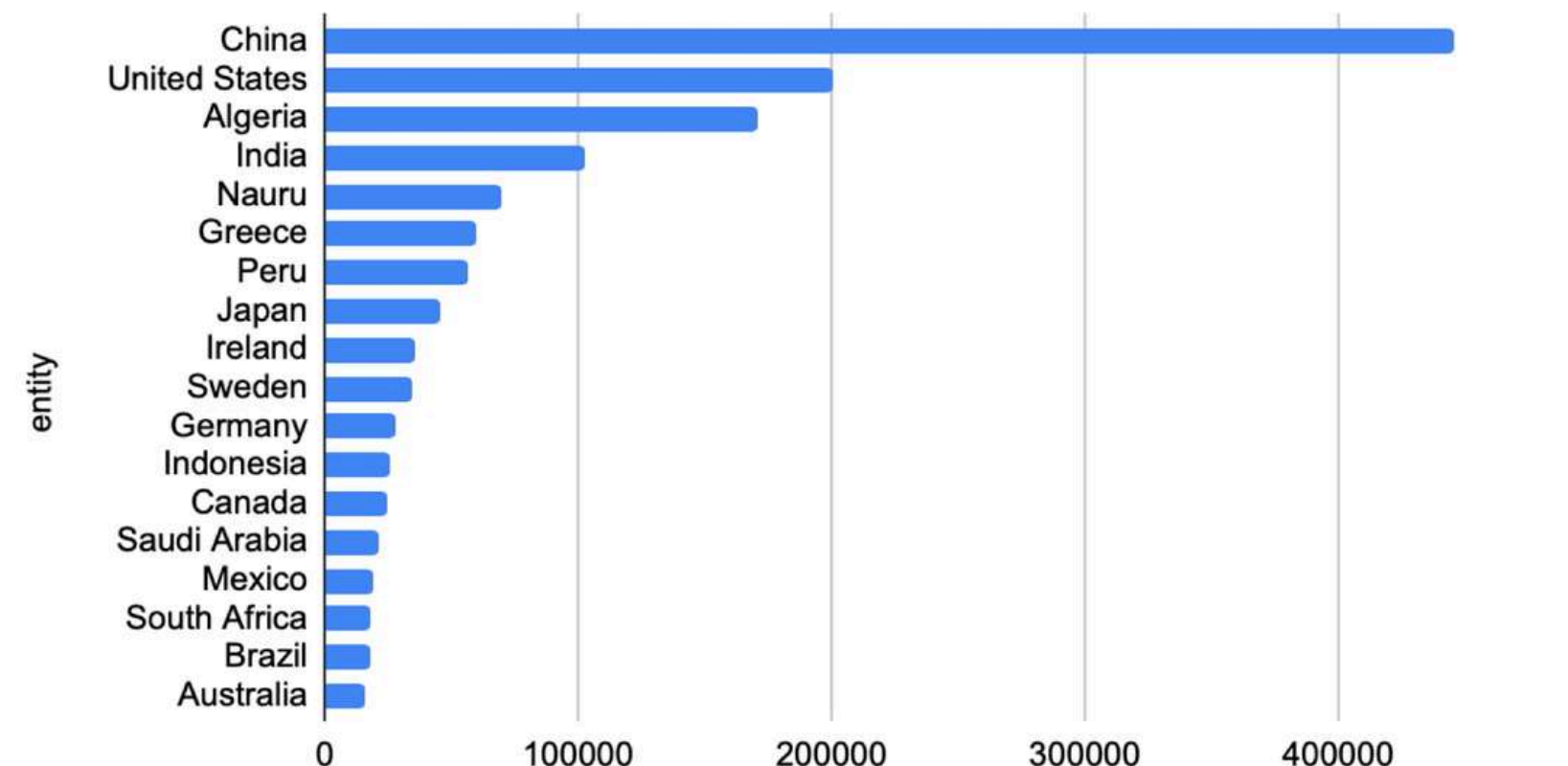
Iceland ranks first, boasting a renewable energy percentage of 80.1% and an average life expectancy of 82.7 years. Next is **Norway**, with 67.1% of energy from renewable sources and a slightly higher life expectancy of 82.8 years. **Brazil**, despite having a significant share of renewable energy at 47.3%, has a lower life expectancy of 75.7 years. **Sweden** reaches 43.8% renewable energy, with a notable life expectancy of 82.5 years. Finally, **New Zealand** stands out with 36.8% renewable energy and a life expectancy of 81.9 years, confirming the balance between sustainability and population well-being.

WHICH COUNTRIES HAD CO2 EMISSIONS ABOVE THE GLOBAL AVERAGE IN 2019?

```
WITH global_avg_co2 AS (  
  SELECT AVG(co2_emissions_kt_by_country) AS avg_co2  
  FROM sustainable_energy  
  WHERE year = '2019'  
)  
SELECT entity, co2_emissions_kt_by_country  
FROM sustainable_energy, global_avg_co2  
WHERE year = '2019' AND co2_emissions_kt_by_country > avg_co2  
ORDER BY co2_emissions_kt_by_country DESC;
```

- **China and the United States** are the largest CO₂ emitters, with values significantly higher than those of other countries.
- **India and Germany** follow with considerable emissions, though lower than China and the United States.
- Countries such as **Brazil and Australia** have lower CO₂ emissions compared to the top four, but still above the global average.

co2_emissions_kt_by_country rispetto a entity



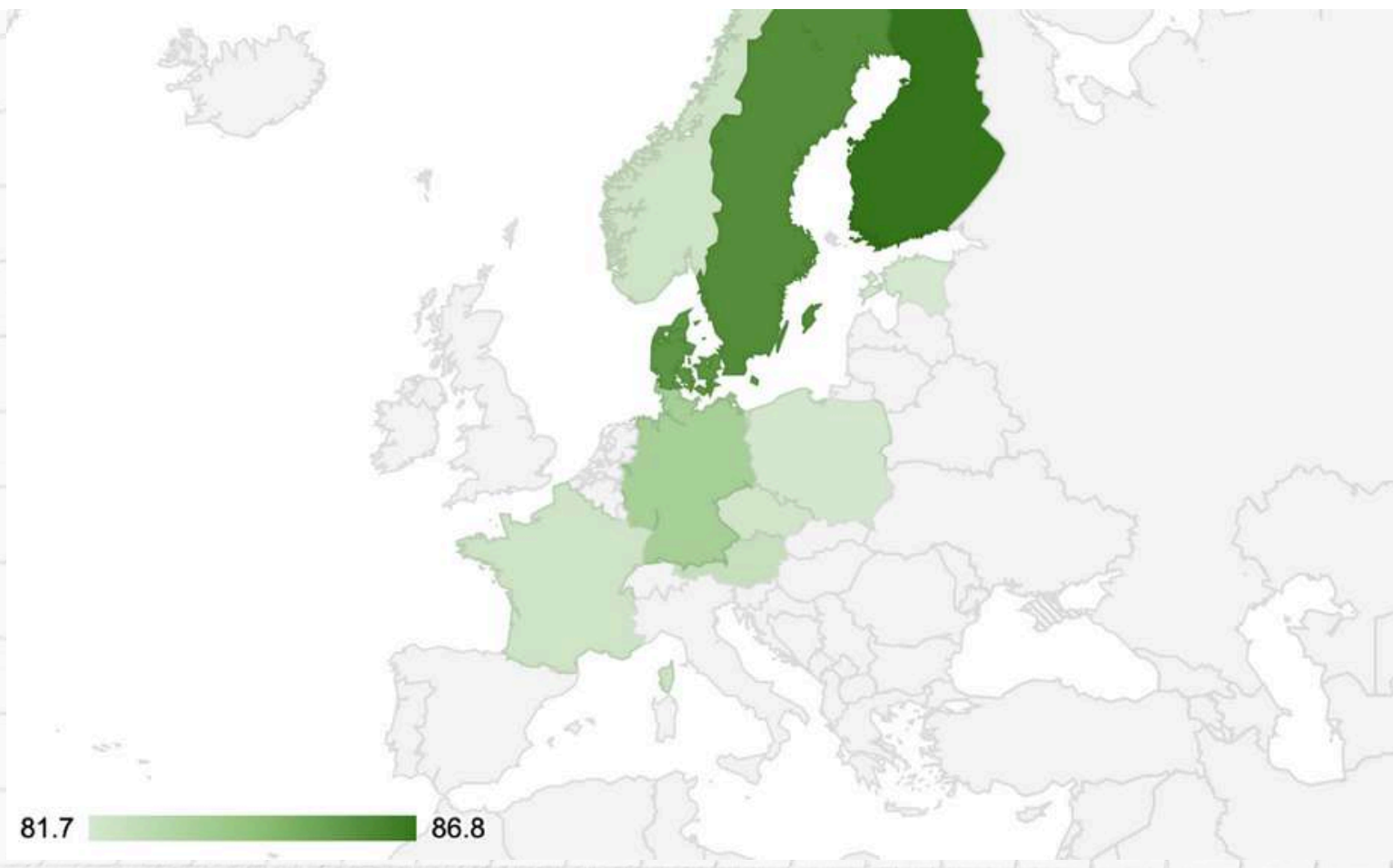
This analysis highlights the importance of focusing on the most polluting countries to reduce global emissions. Emission reduction strategies must be tailored to the specific needs and contexts of each country.

WHICH COUNTRIES HAVE THE HIGHEST OVERALL SDG INDEX IN 2022?

```
SELECT country, sdg_index_score
FROM sdg_index_2000_2022
WHERE year = '2022'
ORDER BY sdg_index_score DESC
LIMIT 10;
```

The analysis shows that in 2022, the countries with the highest overall SDG Index scores are predominantly Nordic, with **Finland** in first place (86.80), followed by **Sweden** and **Denmark**.

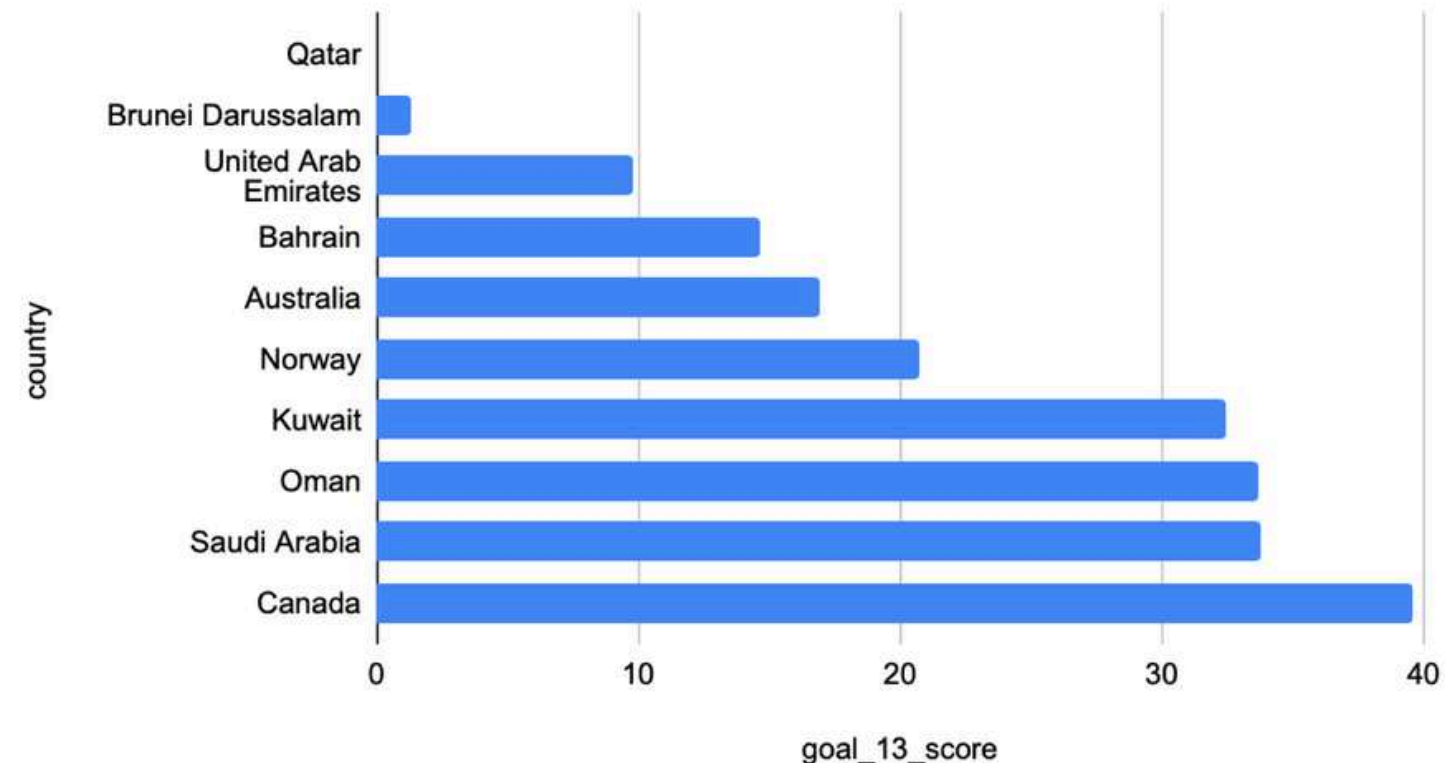
Other European countries, such as **Germany**, **Austria**, and **France**, also achieve high scores, demonstrating a strong commitment to achieving the **Sustainable Development Goals**.



WHICH COUNTRIES HAVE THE LOWEST SCORE FOR GOAL 13 (CLIMATE ACTION) IN 2022?

```
SELECT country, goal_13_score
FROM sdg_index_2000_2022
WHERE year = '2022'
ORDER BY goal_13_score ASC
LIMIT 10;
```

goal_13_score rispetto a country



At the top with the worst score is **Qatar** with a score of 0.00, followed by **Brunei Darussalam** (1.30) and the **United Arab Emirates** (9.80). Other countries with very low scores include **Bahrain** (14.60), **Australia** (16.90), and **Norway** (20.70). Countries like **Kuwait**, **Oman**, **Saudi Arabia**, and **Canada** also show relatively low scores, ranging from 32.40 to 39.60.

These results indicate that many resource-rich countries, particularly those with fossil fuels, have low scores for climate action, reflecting poor performance in addressing emissions and climate adaptation challenges.

COMPARISON OF 2022 SDG SCORES FOR COUNTRIES WITH THE HIGHEST CO2 EMISSIONS

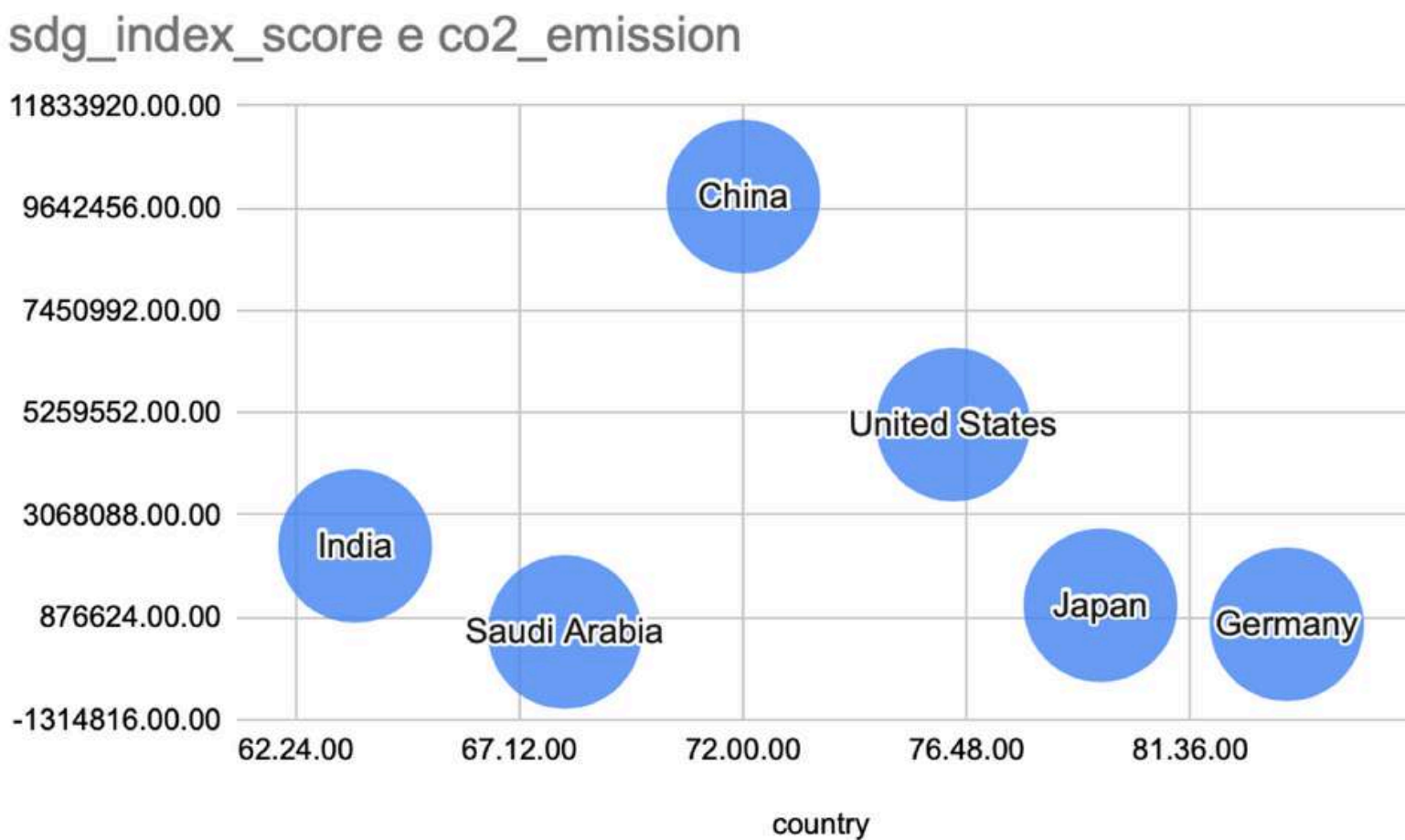
```
WITH top_co2_countries AS (  
  SELECT country  
  FROM country_2023  
  ORDER BY co2_emission DESC  
  LIMIT 10)  
SELECT s.country, s.sdg_index_score, c.co2_emission  
FROM sdg_index_2000_2022 s  
JOIN country_2023 c ON s.country = c.country  
WHERE s.year = '2022' AND s.country IN (SELECT country FROM top_co2_countries)  
ORDER BY s.sdg_index_score DESC;
```

The data show that **Germany** has the highest SDG score (83.40), indicating strong progress toward sustainable development goals, while **India** has the lowest score (63.40).

In terms of CO2 emissions, the **United States** and **China** are the main emitters, reflecting their high level of industrialization.

Germany and Japan have relatively low emissions compared to their economic size, suggesting a stronger commitment to sustainable policies.

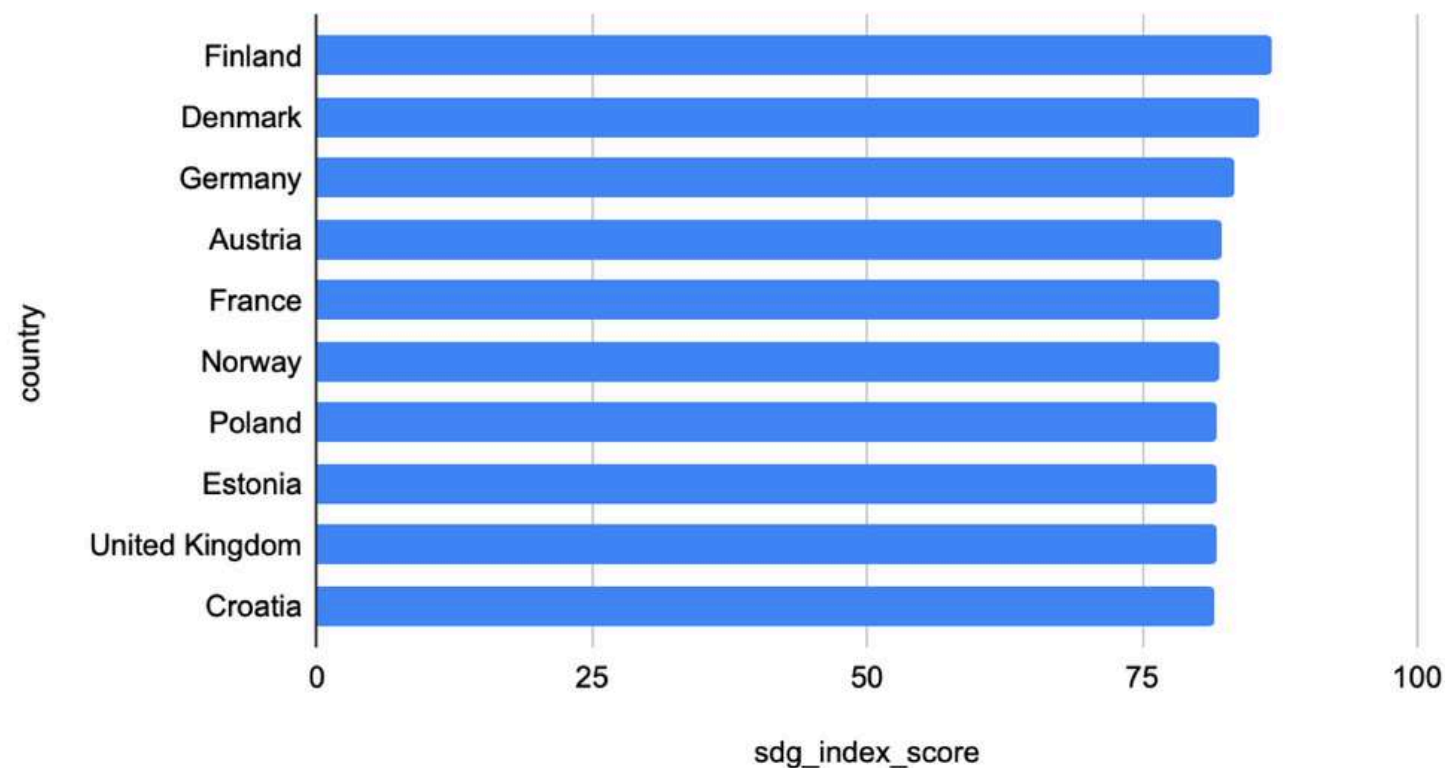
However, there is still a significant **discrepancy** between **sustainable development** and **environmental impacts**.



TOP 10 EUROPEAN COUNTRIES WITH THE HIGHEST SDG_INDEX_SCORE

```
WITH europe AS (  
  SELECT country  
  FROM country_2023  
  WHERE abbreviation IN ('AL', 'AD', 'AM', 'AT', 'BY', 'BE', 'BA', 'BG', 'HR', 'CY', 'CZ',  
    'DK', 'EE', 'FI', 'FR', 'GE', 'DE', 'GR', 'HU', 'IS', 'IE', 'IT', 'KZ', 'XK', 'LV',  
    'LI', 'LT', 'LU', 'MT', 'MD', 'MC', 'ME', 'NL', 'MK', 'NO', 'PL', 'PT', 'RO', 'RU',  
    'SM', 'RS', 'SK', 'SI', 'ES', 'SE', 'CH', 'TR', 'UA', 'GB', 'VA')  
)  
SELECT year, country, sdg_index_score  
FROM sdg_index_2000_2022  
WHERE year = '2022' AND country IN (SELECT country FROM europe)  
ORDER BY sdg_index_score DESC  
LIMIT 10;
```

sdg_index_score rispetto a country



These results show that **Finland** is the country with **the highest SDG score** in Europe, followed by **Denmark** and **Germany**. The presence of Scandinavian countries (**Finland, Denmark, Norway**) in the top 10 confirms their strong commitment to sustainable development, a trend already known due to their progressive environmental, social, and economic policies.

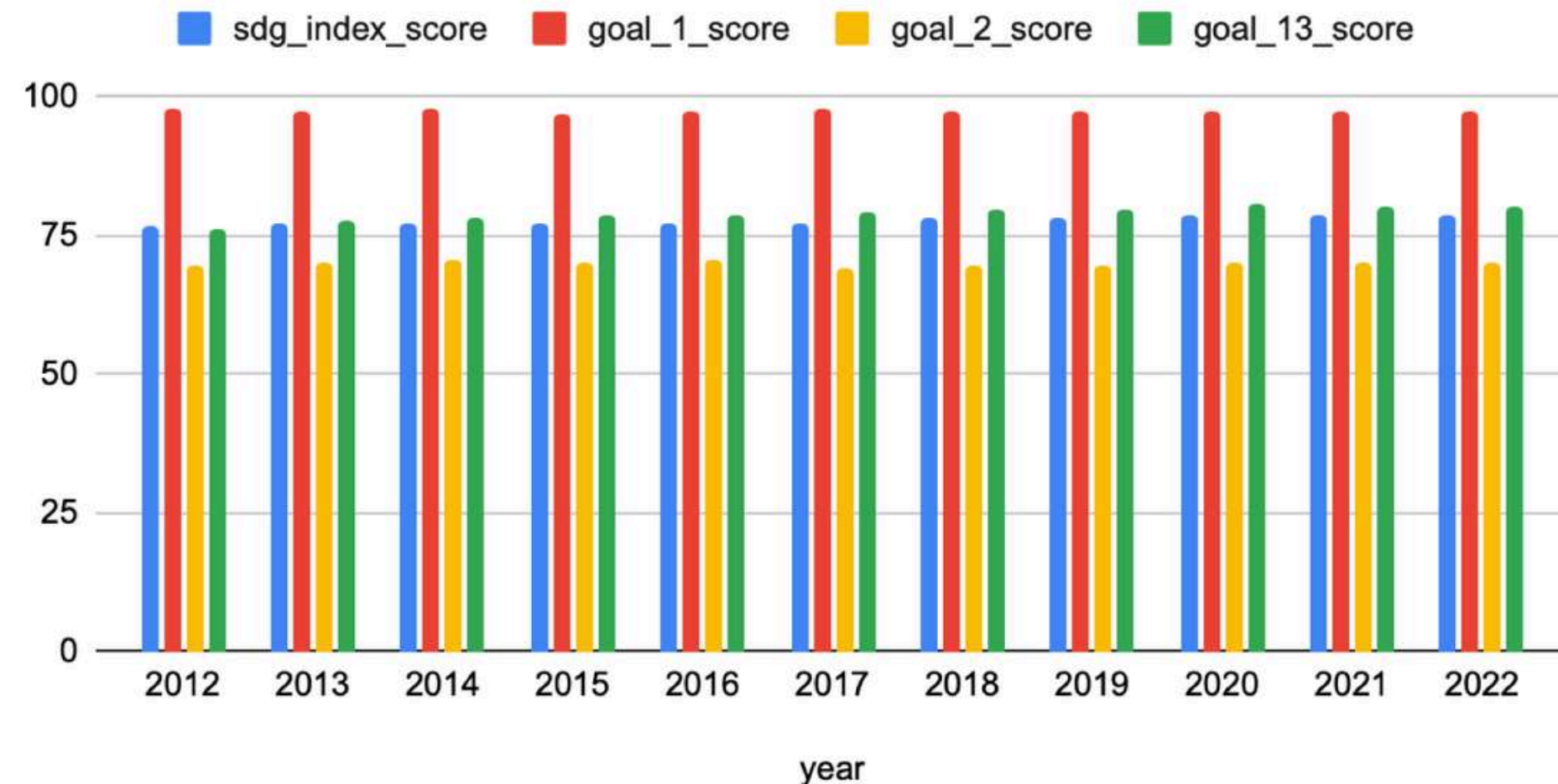
Germany and **France**, two of the largest European economies, also demonstrate strong performance in their SDG objectives. Surprisingly, countries like **Poland, Estonia**, and **Croatia** are included in the top 10, showing significant progress toward sustainable development goals despite coming from a less strong economic and social context compared to other Western European countries.

Overall, European countries show a strong commitment to the **SDGs**, with generally high scores indicating good performance across various aspects of **sustainable development**.

REND OF ITALY'S SDG SCORES FROM 2012 TO 2022

```
SELECT year, sdg_index_score, goal_1_score, goal_2_score, goal_13_score
FROM sdg_index_2000_2022
WHERE country = 'Italy' AND year BETWEEN '2012' AND '2022'
ORDER BY year;
```

sdg_index_score, goal_1_score, goal_2_score e
goal_13_score



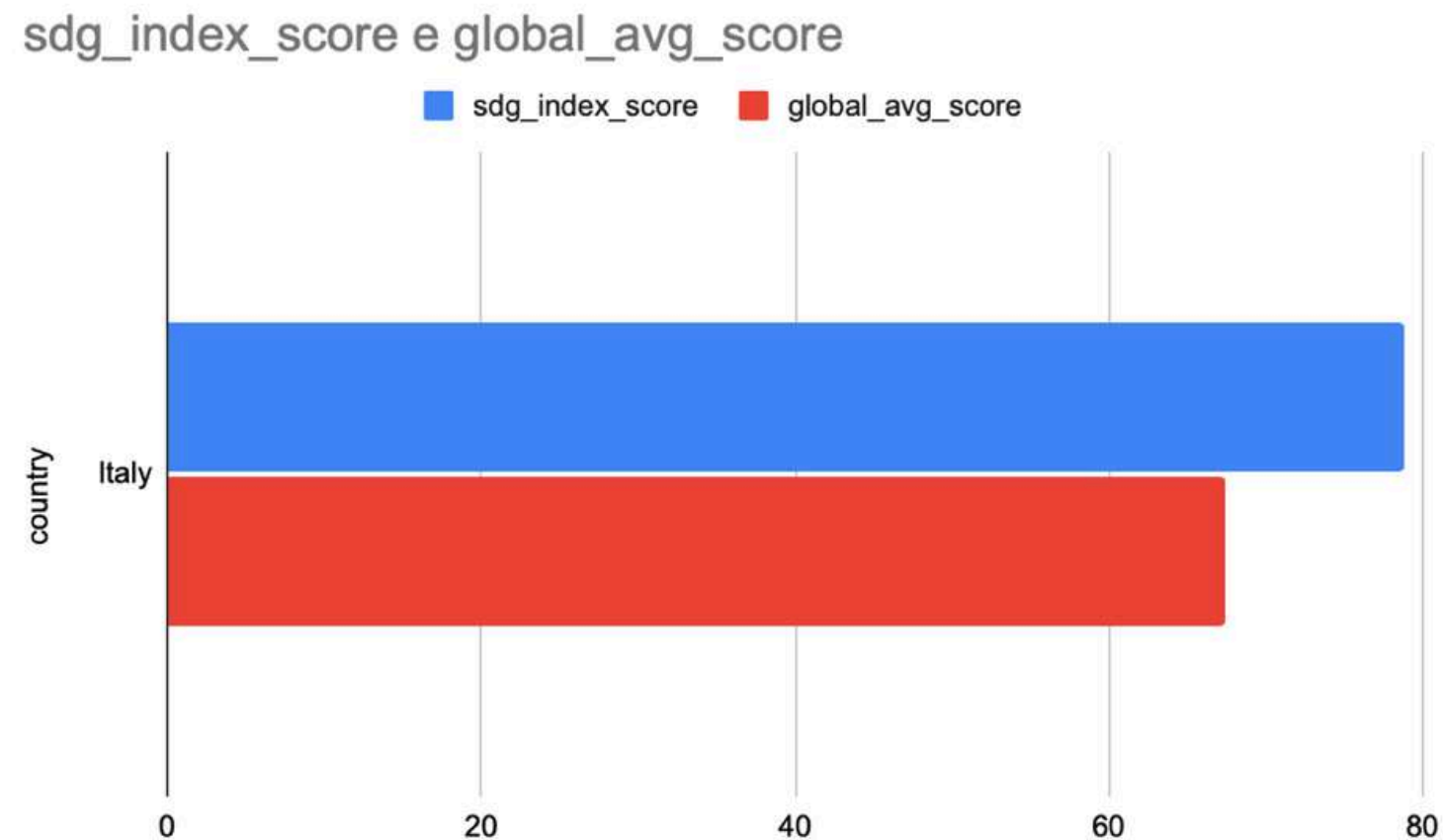
From **2012 to 2022**, Italy's SDG index steadily increased from 76.50 to 78.80. Scores for specific goals (**Goal 1, Goal 2, Goal 13**) remained stable, indicating overall strong performance.

- **Goal 1 (Poverty)**: Score consistently around 97.50, suggesting good performance in poverty reduction.
- **Goal 2 (Zero Hunger)**: Slight increase from 69.50 to 69.80, indicating minimal improvement in combating hunger.
- **Goal 13 (Climate Action)**: Noticeable increase from 76.20 to 79.90, showing significant progress in climate actions.

Italy maintained strong performance on SDG goals from 2012 to 2022, with notable improvements in climate actions, but there is still room for progress, especially regarding hunger-related goals.

COMPARISON BETWEEN ITALY'S SDG SCORE AND THE GLOBAL AVERAGE IN 2022

```
WITH global_avg AS (  
  SELECT AVG(sdg_index_score) AS global_avg_score  
  FROM sdg_index_2000_2022  
  WHERE year = '2022'  
)  
SELECT s.country, s.sdg_index_score, global_avg.global_avg_score  
FROM sdg_index_2000_2022 s, global_avg  
WHERE s.country = 'Italy' AND s.year = '2022';
```



From the obtained data, we can see that **Italy** had an **SDG** score of 78.80 in **2022**, which is significantly higher than **the global average** of 67.37. This indicates that Italy performed better than the global average in achieving the **Sustainable Development Goals (SDGs)**.

This result reflects a consistent commitment and significant progress in meeting the SDGs.

CONCLUSION

The results highlighted that European countries, especially Nordic ones like **Finland** and **Sweden**, achieved the highest scores in the **SDG Index**, demonstrating a strong commitment to sustainability.

In contrast, resource-rich countries, such as **Qatar**, showed significant gaps in addressing climate change.

Additionally, **China** and the **United States** emerged as the largest contributors to **global CO2 emissions**, emphasizing the need for stricter policies.

Italy demonstrated good progress toward sustainable development goals, surpassing the global average in 2022.

Overall, the data reveal a complex picture where global sustainability efforts are uneven and need to be further intensified to tackle environmental challenges.





THANK YOU!

[CLICK HERE TO ACCESS THE SQL
SCRIPT AND PROJECT PRESENTATION](#)

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